Jinpeng Lu

jinplu@mail.ustc.edu.cn Website Scholar

Video World Models | Generative Video Evaluation | Embodied AI

Research Profile & Core Strengths

My research focuses on **video world models and generative video evaluation**, particularly world-state persistence under viewpoint changes. I also study **embodied AI and vision-language-action (VLA) models**, with an exploratory interest in the verifiability, safety, and evaluation of **recursive self-improvement (RSI)**.

First-author work includes WRBench world-state evaluation and VeloxSeg / H2ASeg for efficient 3D medical perception. I also co-authored VLA-Trace and Pelican-Unify 1.0.

My methods span diagnostic benchmark design, representation and behavior analysis, efficient 3D modeling, and reproducible PyTorch / DDP / CUDA training, evaluation, and ablation pipelines.

Education

University of Science and Technology of China Sep. 2025 – Present
M.S. in Information and Communication Engineering, expected Jun. 2028
Advisor: Zhiwei Xiong; interests: video world models, generative video evaluation, embodied AI / VLA models; exploratory interest in verifiability, safety, and evaluation of RSI.

Harbin Institute of Technology, Shenzhen Sep. 2021 – Jun. 2025
B.S. in Data Science and Big Data Technology (Mathematics discipline); GPA: 3.728/4, Rank: 17/172, Top 10%
Core training: mathematics, statistics, numerical computing, data engineering; research training: PET/CT 3D segmentation, generative signal enhancement.

Representative Works

Video World Models & Generative Video Evaluation

Current World Models Lack a Persistent State Core
arXiv preprint, First Author **2026.06**
- **Question & Approach:** Asked whether video generation models preserve evolving state after a target leaves and re-enters view; built **WRBench** to control observability through camera motion.
- **Evidence & Result:** Evaluated **23 models**, **four control paradigms**, and **9,600 videos**, calibrated with **2,547 human judgments**; visual quality and successful re-observation did not reliably imply correct event-state consistency.

Embodied Intelligence & VLA

VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing
arXiv preprint, Co-author **2026.05**
- **Question & Approach:** Combined representation alignment and drift analysis, causal attention interventions, and behavioral probes to trace VLA models from internal states to decisions and embodied rollouts.
- **Finding:** Experiments on $\pi_{0.5}$ and OpenVLA identified distinct adaptation and action-routing patterns while revealing limits in fine-grained semantic following.

Medical AI

VeloxSeg: JL-Guided Efficient 3D Medical Segmentation
ICLR 2026, First Author [CCF-A] **2026.04**
- **Method:** Designed a lightweight CNN-Transformer with multi-scale cross-modal context, JL-guided local 3D features, and training-time texture-prior transfer.
- **Result:** Reached **62.51 / 56.48 Dice** on AutoPET-II / Hecktor2022 with **1.66M parameters**, improving GPU/CPU efficiency and memory use.

H2ASeg: Hierarchical PET/CT Tumor Segmentation
MICCAI 2024, First Author [CCF-B] **2024.10**
- **Method:** Framed 3D PET/CT tumor segmentation as multimodal volumetric perception and built hierarchical local/global cross-modal interaction with target-aware feature reweighting.
- **Result:** Reached **60.03 / 59.69 Dice** on AutoPET-II / Hecktor2022; best-baseline gains: **+2.70 pp / +4.9 pp**.

Core Methods & Engineering

Research: Video world models, generative video evaluation, world-state persistence, embodied AI and VLA analysis; medical AI, PET/CT and MRI 3D segmentation
Engineering: Python, PyTorch, CUDA, DDP, Shell/Bash, LaTeX; training/evaluation pipelines, experiment reproduction, ablation studies, custom model implementation
Tools: Linux, Git, Docker, Weights & Biases / TensorBoard, vision, multimodal, and medical-imaging data-processing toolchains
Writing: English manuscript writing for ICLR / MICCAI / Medical Image Analysis submissions, experiment analysis, and reproducibility documentation

Full Research Outputs

Video World Models & Generative Video **2026 – Present**

[1] **Current World Models Lack a Persistent State Core**, *First Author* **arXiv 2026**

Embodied AI & VLA **2026 – Present**

[1] **VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing**, *Co-author* **arXiv 2026**

[2] **Pelican-Unify 1.0: A Unified Embodied Intelligence Model**, *Co-author* **arXiv 2026**

Medical AI **2023 – 2026**

[1] **Johnson-Lindenstrauss Lemma Guided Network for Efficient 3D Medical Segmentation**, *First Author* **ICLR 2026 [CCF-A]**

[2] **H2ASeg: Hierarchical Adaptive Interaction and Weighting Network for Tumor Segmentation in PET/CT Images**, *First Author* **MICCAI 2024 [CCF-B]**

[3] **Does DINOv3 Set a New Medical Vision Standard?**, *Co-first Author* **arXiv 2025**

[4] **AttriMIL: Revisiting Attention-based Multiple Instance Learning for WSI Classification**, *Co-author* **Medical Image Analysis 2025 [CCF-C] [SCI Q1]**

[5] **IPGPhormer: Interpretable Pathology Graph-Transformer for Survival Analysis**, *Co-author* **arXiv 2025**

[6] **A Localization-to-Segmentation Framework for Automatic Tumor Segmentation in Whole-Body PET/CT Images**, *Co-author* **arXiv 2023**

Additional Project

Generative Sensor Signal Enhancement Jun. 2023 – Jun. 2024

Lead, National Undergraduate Innovation Program

- Reproduced and compared GAN, diffusion, and flow-based models; designed conditional guidance strategies to improve reconstruction quality and perceptual fidelity, producing a reproducible comparison pipeline and experiment report.